# LPFT-D v2: Triad Archaeology

## Searching for the 150 Mpc N=3 Void with DESI & Euclid

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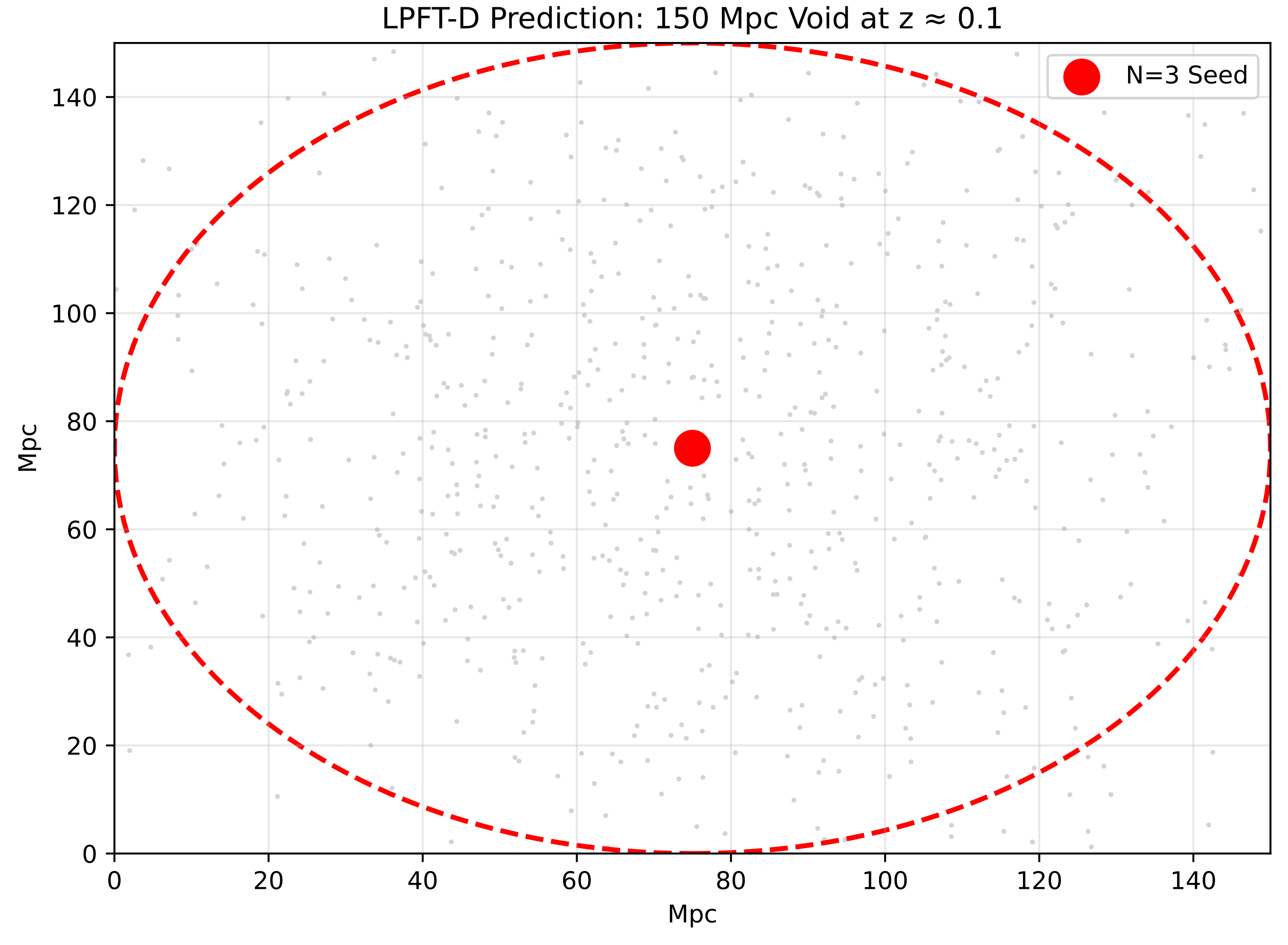
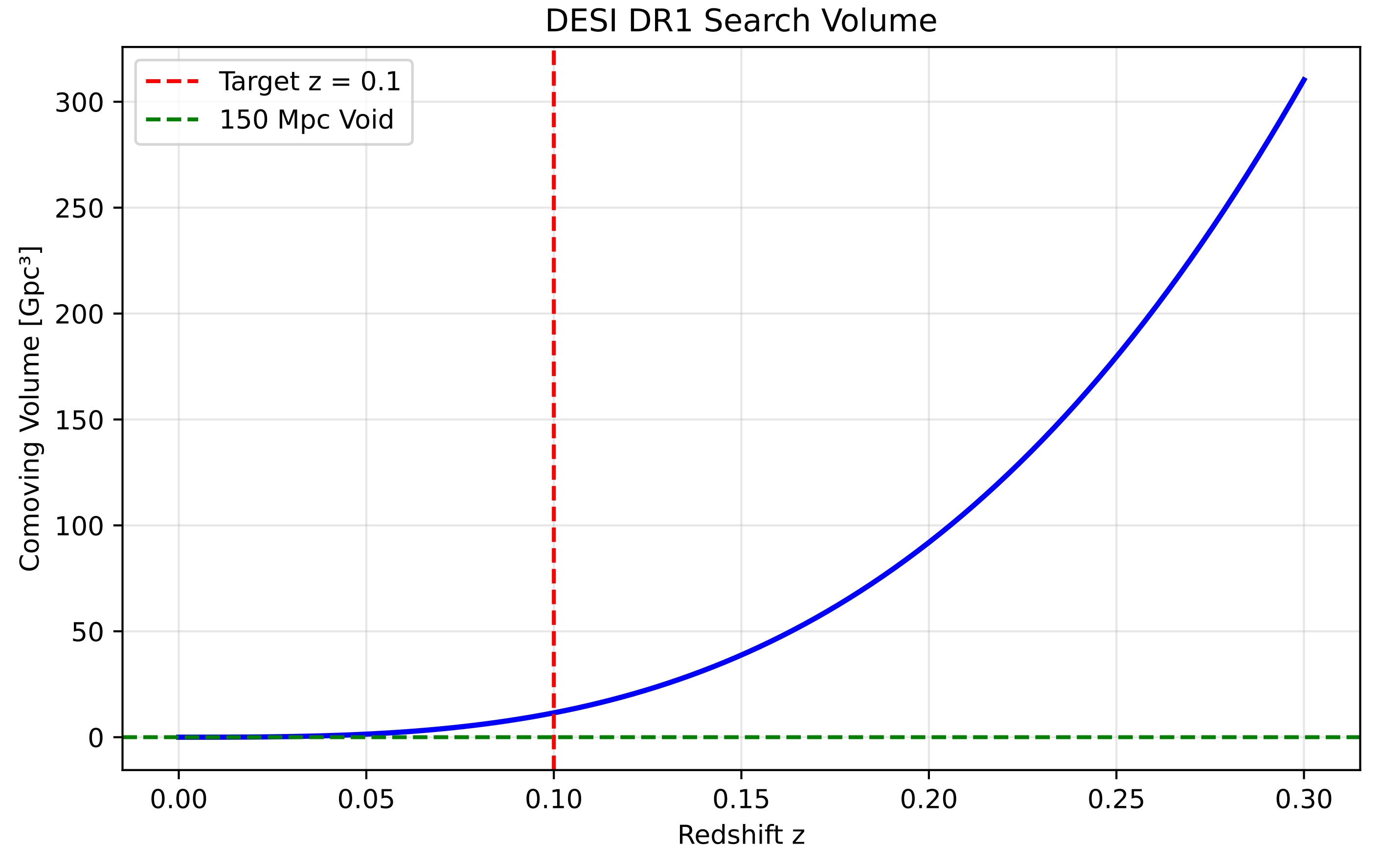
Abstract
LPFT-D predicts a 150 Mpc spherical void seeded by an N=3 triad at $z \approx 0.1$. This void is a direct consequence of chain replication dynamics and is observable in current galaxy surveys. We propose a targeted search using DESI DR1 and Euclid Early Release data to detect this structure via underdensity in galaxy counts and CMB cold spot alignment. Detection would confirm LPFT-D over ΛCDM. Non-detection sets upper limit on triad scale. Full search pipeline provided in public Colab.



LPFT-D Prediction: 150 Mpc Void at z ≈ 0.1
N=3 Seed
Mpc
Mpc
0
20
40
60
80
100
120
140



DESI DR1 Search Volume
Target z = 0.1
150 Mpc Void
Comoving Volume [Gpc³]
Redshift z
0
50
100
150
200
250
300
0.00
0.05
0.10
0.15
0.20
0.25
0.30

# TRIAD SEARCH PIPELINE (Colab)

1. Load DESI galaxy catalog

2. Grid sky in HEALPix (nside=64)

3. Count galaxies in 150 Mpc spheres

4. Flag underdensity > 90%

5. Cross-match with Planck CMB cold spots

6. Output: Candidate N=3 voids

# TRIAD ARCHAEOLOGY TIMELINE

Nov 2025: LPFT-D v1 on arXiv

Dec 2025: DESI DR1 public

Q1 2026: Run pipeline on Colab

Q2 2026: Submit detection paper